

Covert-ness-Oriented Water-to-Air Optical Wireless Semantic Communications

Qingqing Hu^{ID}, *Member, IEEE*, Bin Chen^{ID}, *Senior Member, IEEE*, Yi Lei^{ID},
and Chen Gong^{ID}, *Senior Member, IEEE*

Abstract—Optical wireless communication (OWC) enables high-rate and low-latency direct data transmission between water and air, whereas wave refraction and particle scattering give rise to the issue of information leakage. To improve the energy efficiency and security in transmission, we propose a covert-ness-oriented semantic communication scheme for water-to-air (W2A) OWC systems in this paper. Based on a theoretical derivation regarding the monotonic correlation between the sparsity ratio of transmission encoding vectors and covert performance metrics, we introduce a sparsity-regularized semantic transmission scheme that extracts task-relevant semantic features, which significantly reduces the transmitted data volume and thereby lowers the probability of being detected by adversarial listeners. Aimed at underwater monitoring tasks, we establish a dedicated dataset with comprehensive coverage of core underwater anomalous object categories: the underwater anomaly monitoring (UAM) dataset. Based on this dataset, we perform thorough training and optimization on the parameters of the proposed scheme and test the scheme effectiveness. Simulation results demonstrate that the proposed scheme achieves high accuracy in anomaly detection under various noise conditions, while significantly improving signal covert-ness via adaptive sparsification. The related code and dataset will be available at <https://github.com/qingqinghu-Ricky/Covert-semantic-communication-for-W2A-OWC> upon publication.

Index Terms—Optical wireless communication, semantic communication, covert transmission, water-to-air.

I. INTRODUCTION

WITH the growing demands for underwater resource exploration and air-sea collaborative defense, it is imperative to establish direct and secure cross-medium communication between water and air [1]. Optical wave emerges

as a highly feasible carrier for such direct communication, owing to its high bandwidth, superior flexibility, and relatively moderate underwater propagation loss [2]. However, due to the random refraction of waves and the scattering of underwater particles, optical signals may be subject to the risk of being detected and intercepted by the adversary [3]. Therefore, covert transmission is required to hide the existence of communication behavior and effectively enhances the system security [4].

Recent studies on covert transmission can be divided into several categories. Multiple-antenna beamforming strategies have been proposed to enhance covert-ness in multi-input multi-output (MIMO) [5], millimeter-wave [6], and frequency-diverse-array-based near-field communication systems [7]. In both uplink and satellite non-orthogonal multiple access (NOMA) systems, covert transmitters leveraged public message broadcasts from reliable users as a cover to transmit their private information [8]. Artificial noise (AN) or jamming signals were generated by multi-antenna transmitters [9], friendly jammers [10], and full-duplex (FD) receivers [11] to assist in covert signal transmission. Intelligent reflecting surfaces (IRSs) were adopted to enable the amplification of received signal power at legitimate receivers and the suppression of received power at adversary listeners in [12]. Multiple system transmission parameters, including the channel selection probability [13], transmission power [14], flying trajectories [15], are optimized to satisfy the covert-ness constraints. In particular, for water-to-air (W2A) optical wireless communication (OWC) systems, prior work [16] have resorted to strong background light to facilitate the covert transmission.

However, the aforementioned covert schemes neglect the intrinsic meaning of transmitted messages and require various auxiliary resources and conditions to transmit the complete content of messages, which fundamentally undermines the general applicability of practical deployment. To address this limitation, semantic communication emerges as a paradigm-shifting communication framework that fundamentally differs from Shannon's conventional content-delivery communication paradigm. Conventional communication systems are designed based on the bit-level transmission principle, which treat all data bits as equally valuable and aim to achieve error-free recovery of the entire bit stream at the receiver, with bit error rate (BER) and symbol error rate (SER) as the core performance metrics [17]. In contrast, semantic communication is a task-oriented communication paradigm whose core objective is

Received 11 February 2026; revised 9 May 2026 and 13 July 2026; accepted 11 September 2026. Date of publication 16 September 2026; date of current version 22 September 2026. The work of Qingqing Hu was supported in part by the National Natural Science Foundation of China (NSFC) under Grant 62501223 and in part by the Fundamental Research Funds for the Central Universities under Grant JZ2025HGTA0158. The work of Bin Chen was supported in part by NSFC under Grant 62571176 and in part by Anhui Provincial Natural Science Foundation under Grant 2508085J041. The work of Chen Gong was supported by NSFC under Grant 62331024. The editor coordinating the review of this article was J. Huang. (*Corresponding author: Bin Chen.*)

Qingqing Hu, Bin Chen, and Yi Lei are with the School of Computer Science and Information Engineering, Hefei University of Technology, Hefei 230002, China (e-mail: qingqinghu@hfut.edu.cn; bin.chen@hfut.edu.cn; leiyi@hfut.edu.cn).

Chen Gong is with CAS Key Laboratory of Wireless-Optical Communications, University of Science and Technology of China, Hefei 230026, China (e-mail: cgong821@ustc.edu.cn).

Digital Object Identifier 10.1109/TGCN.2026.3734354

2473-2400 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: New York University Abu Dhabi Campus. Downloaded on October 02, 2026 at 05:42:13 UTC from IEEE Xplore. Restrictions apply.

TABLE I
LITERATURE COMPARISON ON COVERT SEMANTIC COMMUNICATION

Ref.	System	Data type	Covert strategy		Semantic communication model	
[20]	RF	Text	Additional auxiliary resource	FD-receiver-based AN	Semantic-only networks	DeepSC from [22]
[21]	RF	Text		FD-receiver-based AN		VQ-VAE from [23]
[24]	RF	Image		Jammer		Resnet from [26]
[25]	RF	Image		Jammer&Time chosen		SGG from [27]
Ours	W2A-OWC	Image	Sparsity-regularization-based semantic network motivated by a covertness analysis			

to accurately reconstruct the semantic meaning of information (i.e., the utility of information for a specific target task) rather than the exact bit sequence [18]. Unlike conventional systems that transmit full original data via unified coding and modulation without distinguishing the semantic value of different bits, semantic communication leverages deep learning to hierarchically extract only task-critical semantic features and filter out redundant information [19]. This differentiated processing logic directly translates to superior resource utilization efficiency: conventional communication allocates equal transmission resources to all bits, leading to poor performance in bandwidth- and energy-constrained scenarios, while semantic communication concentrates resources on high-value semantic information to achieve significantly higher energy and spectral efficiency. More importantly, the drastic reduction in transmitted symbols through semantic compression inherently enhances communication covertness, as large data volumes of conventional systems result in long signal durations and distinct power signatures that are easily detected by adversaries. Compared with current content-delivery-type communication, semantic communication is particularly suitable for W2A-OWC scenarios, where the cross-medium channels suffer from complex propagation losses and information leakage risks. By transmitting only core semantic information tailored to diverse task requirements, semantic communication significantly reduces the data volume and transfer time, ultimately improving the energy efficiency and lowering the probability of detection by adversarial listeners.

In recent years, several methods and technologies have been studied for developing covertness-oriented semantic communication systems, as shown in Table I. Prior works such as [20] and [21] adopted AN emitted by a FD receiver to assist in disturbing the adversary's detection, where the semantic information of text data was extracted by the existing models from [22] and [23]. In [24], a covert semantic communication framework tailored for image data was proposed by leveraging a friendly jammer to transmit jamming signals toward the vulnerable device. In addition to deploying a friendly jammer, paper [25] strategically selected transmission time slots and power to maximize the covertness and accuracy of semantic information. However, the aforementioned schemes in radio frequency (RF) systems all relied on the assistance of additional jammers or FD receivers for covert transmission, making them inapplicable to resource-constrained scenarios. Furthermore, these schemes employed pre-existing network architectures for semantic information extraction, without considering the joint optimization of semantics and covertness.

Despite significant advancements in underwater optical semantic communication, existing works in [28], [29] and [30] have concentrated on image reconstruction, with performance evaluation confined to conventional image quality metrics including peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and learned perceptual image patch similarity (LPIPS). To address this research gap, this paper proposes a novel covertness-oriented semantic communication scheme for W2A-OWC systems, targeting semantic segmentation for underwater anomaly monitoring applications. The main contributions of this work are summarized as follows:

- 1) We state the KL-divergence monotonicity induced by sparse signaling. Based on the monotonicity, we propose a covertness-oriented semantic communication scheme for W2A-OWC systems. The proposed scheme introduces a semantic transmission architecture with a sparse mask generator, enabling the integrated design and optimization of core semantic information extraction and a noise-adaptive sparse signal generation, without the need for additional auxiliary resources, such as FD receivers and jammers.
- 2) Targeting underwater monitoring tasks in W2A-OWC systems, we construct a dedicated dataset encompassing diverse categories of underwater anomalous objects for the first time. The proposed dataset provides sufficient pixel-level annotations for three broad classes of underwater anomalous objects, addressing the limitations of existing underwater semantic segmentation datasets in security-relevant anomalous object sample size.
- 3) We test the performance of proposed scheme in underwater anomaly monitoring tasks. Simulation results indicate that the proposed scheme achieves high semantic transmission accuracy under different channel noise conditions and enables adaptive adjustment of the sparsity ratio, thereby significantly reducing the transmitted data volume and enhancing the signal covertness.

The remainder of this paper is organized as follows. Section II introduces the W2A-OWC system model and theoretically derives the relationships between the sparsity ratio of transmission encoding vectors and covert performance metrics. Section III presents an underwater semantic segmentation dataset for anomalous objects, and proposes a covertness-oriented semantic communication scheme for W2A-OWC systems. Section IV evaluates the performance of proposed scheme under varying channel noise conditions for underwater monitoring tasks. Section V concludes this work.

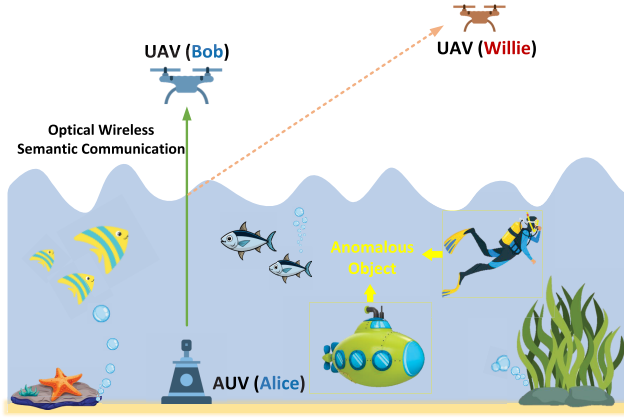


Fig. 1. Illustration of a W2A-OWC system.

II. SYSTEM MODEL

A. System Model

Consider a W2A-OWC system as shown in Fig. 1. An autonomous underwater vehicle (AUV) (legitimate transmitter Alice) adopts optical wave to transmit semantic information of anomalous objects to an unmanned aerial vehicle (UAV) (legitimate receiver Bob) in the air, while another UAV (adversary Willie) tries to detect whether the transmission from Alice to Bob happens. Due to the random fluctuation of the water surface, the channel gains from Alice to both receivers vary randomly with time, modeled as lognormal block fading from experimental measurements [31]. Let x denote the transmitted electrical-domain symbol. After the electrical-to-optical (E/O) conversion of the light-emitting diode (LED), the transmitted optical signal is

$$P = \eta_l (x + I_{dc}), \quad I_{dc} \geq |x|, \quad \mathbb{E}[P] \leq P_{\text{avg}}, \quad (1)$$

where I_{dc} is the direct current (DC) bias; P_{avg} is the average optical power constraint; η_l is E/O conversion efficiency. In the receiver, the electrical-domain signal from the photodiode (PD) after DC blocking can be expressed as

$$y_k = \eta_{p_k} h_k P - \mathbb{E}[\eta_{p_k} h_k P] + n_k = \eta_{p_k} \eta_l h_k x + n_k = \eta_k h_k x + n_k, \quad (2)$$

where $k \in \{b, w\}$ represents Bob and Willie, respectively; η_{p_k} is the optical-to-electrical (O/E) conversion efficiency; $\eta_k = \eta_l \eta_{p_k}$ is overall efficiency; h_k is the W2A-OWC channel gain following a lognormal distribution with parameters μ_k and σ_k ; n_k is modeled as the additive white Gaussian noise (AWGN) with zero mean and variance N_k , since we consider a conventional intensity modulation/direct detection (IM/DD) waveform receiver rather than a photon-counting receiver, and the Poisson shot-noise process is commonly approximated by a Gaussian random variable in this case [32], which is widely used in the noise model of OWC systems [33].

B. Symbol Sparsification for Covert Transmission

In a W2A-OWC system, Willie detects whether Alice transmits signals to Bob based on hypothesis test in each time block, where $\mathcal{H}_0$ denotes the hypothesis of no transmission,

and $\mathcal{H}_1$ denotes the hypothesis of transmission. Consider the block fading, i.e., channel gain h_w remains constant within one time block and changes independently and randomly between blocks, the received electrical signal of Willie in the i -th symbol duration is given by

$$y_w(i) = \begin{cases} n_{w0}(i), & \mathcal{H}_0, \\ \eta_w h_w x(i) + n_{w1}(i), & \mathcal{H}_1, \end{cases} \quad (3)$$

where $n_{w0}(i)$ and $n_{w1}(i)$ are AWGNs under hypothesis $\mathcal{H}_0$ and $\mathcal{H}_1$, respectively.

Assume that the observations $\{y_w(i)\}_{i=1}^{N_t}$ within a block in Eq. (3) are conditionally independent and identically distributed random variables given the channel gain h_w , following $\mathbf{P}_0$ under $\mathcal{H}_0$ or $\mathbf{P}_1$ under $\mathcal{H}_1$, where N_t is the transmitted symbol blocklength. The Kullback-Leibler (KL) divergence $\mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_1)$ is used as a covertness characterization because it has a direct hypothesis-testing interpretation. Although the exact minimum detection-error sum is determined by the total variation distance, total variation is difficult to evaluate for the product distributions observed by Willie. Following the classical low-probability-of-detection analysis of Bash's work [34], KL divergence provides a tractable alternative: by the Chernoff-Stein Lemma [35], under a fixed false-alarm constraint, the optimal missed-detection exponent of Willie's Neyman-Pearson test is determined by $-\mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_1)$. It indicates that reducing $\mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_1)$ slows the exponential decay of Willie's optimal missed-detection probability and directly limits his asymptotic detection capability. Therefore, the steganography and covert communication communities often conclude their proofs by showing an upper bound on the $\mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_1)$ [36], [37].

Let $\mathbf{Q}$ denote the observation distribution at Willie by an arbitrary non-zero transmitted signal. With sparsity ratio $s \in [0, 1]$ (i.e., the probability of the transmitting zero), Willie's observation distribution under the transmission hypothesis $\mathcal{H}_1$ can be written as

$$\mathbf{P}_{1,s} = s\mathbf{P}_0 + (1-s)\mathbf{Q}. \quad (4)$$

Remark 1 (Basic KL Monotonicity Under Sparse Signaling): Consider two sparsity ratios $0 \leq s_1 < s_2 \leq 1$. The corresponding distributions satisfy

$$\mathbf{P}_{1,s_2} = \lambda \mathbf{P}_0 + (1-\lambda) \mathbf{P}_{1,s_1}, \quad \lambda = \frac{s_2 - s_1}{1 - s_1}. \quad (5)$$

Since $0 < \lambda \leq 1$ and KL divergence is convex in its second argument, we have

$$\begin{aligned} \mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_{1,s_2}) &= \mathcal{D}(\mathbf{P}_0 \parallel \lambda \mathbf{P}_0 + (1-\lambda) \mathbf{P}_{1,s_1}) \\ &\leq \lambda \mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_0) + (1-\lambda) \mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_{1,s_1}) \\ &= \frac{1-s_2}{1-s_1} \mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_{1,s_1}) \\ &< \mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_{1,s_1}). \end{aligned} \quad (6)$$

Therefore, Willie's KL divergence $\mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_{1,s})$ is monotonically decreasing as the sparsity ratio s increases. Based on this property, the sparsification design of the transmitted symbols can reduce the distribution discrepancy between the cases of signal transmission and non-transmission, thus enhancing the system covertness.

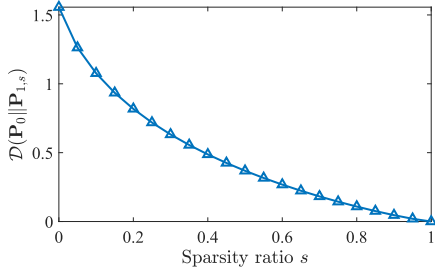


Fig. 2. The monte carlo simulation result of $\mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_{1,s})$.

The Monte Carlo simulation result of $\mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_{1,s})$ under different sparsity ratio s is shown in Fig. 2. The transmitted non-zero signal is assumed to be 4-PAM modulation with pulse amplitudes of 1, 2, 3, and 4 respectively. Each pulse amplitude is transmitted with equal probability, and the noise variance is 1. Monte Carlo results are obtained via the generation of 10^7 samples from the distribution $\mathbf{P}_0$. It is seen that $\mathcal{D}(\mathbf{P}_0 \parallel \mathbf{P}_{1,s})$ decreases monotonically as the sparsity ratio increases, which verifies the correctness of the theoretical derivation.

In the following semantic communication scheme, the transmitted symbols are not generated from a fixed modulation distribution. Instead, the transmitted vector is

$$\mathbf{x}(I) = \mathbf{m}(I) \odot \mathbf{e}(I), \quad (7)$$

where $I \sim \mathcal{P}_{\mathcal{I}}$ denotes an input image drawn from the data distribution, $\mathbf{e}(I)$ is the encoded latent vector, $\mathbf{m}(I)$ denotes the binary sparse mask which determines the sparsity ratio s , $\odot$ denotes element-wise multiplication. Let $\mathbf{P}_{0,h_w}$ denote Willie's observation distribution under $\mathcal{H}_0$ and channel gain h_w . Conditioned on an input image I , the transmitted vector $\mathbf{x}(I)$ and channel gain h_w , Willie's observation distribution under $\mathcal{H}_1$ is denoted by $\mathbf{P}_{y_w|\mathbf{x}(I), \mathcal{H}_1, h_w}$ according to Eq. (3). The distribution under $\mathcal{H}_1$ observed by Willie is the image-induced mixture:

$$\mathbf{P}_{1,h_w} = \mathbb{E}_{I \sim \mathcal{P}_{\mathcal{I}}} [\mathbf{P}_{y_w|\mathbf{x}(I), \mathcal{H}_1, h_w}]. \quad (8)$$

The average KL divergence in the following simulations is calculated over test images and then averaged over sampled block-fading coefficients, i.e., $\mathbb{E}_{h_w} [\mathcal{D}(\mathbf{P}_{0,h_w} \parallel \mathbf{P}_{1,h_w})]$.

III. COVERTNESS-ORIENTED SEMANTIC COMMUNICATION SCHEME

A. The Underwater Anomaly Monitoring (UAM) Dataset

Nowadays, there are lots of underwater semantic segmentation datasets. The SUIM dataset proposed in [38] contains 1525 training images and 110 test images in total, among which only 506 images include anomalous objects (human divers and robots); while the COU dataset in [39] provides 10K instance-segmented man-made underwater objects, the vast majority of these samples are common underwater debris (e.g., plastic bags, bottles, and lures), with fewer than 300 valid images of core anomalous objects (remotely operated vehicles (ROVs) and AUVs) central to underwater security and monitoring applications. The LIACI dataset in [40] is exclusively designed for underwater ship hull inspection, comprising 1893

images annotated with 10 ship-related categories including corrosion, paint peel, and marine growth, but it does not cover any of the general anomalous objects. These severe limitations in anomalous object sample size, category coverage, and task applicability across existing datasets motivate us to construct the UAM dataset, a dedicated dataset specifically tailored for general underwater anomalous object segmentation tasks. The UAM dataset covers various anomalous object categories including AUV, ship, submarine, diver and others, which comprises 4298 RGB images for training and validation, and 1075 RGB images for testing.

These images are carefully collected from the Internet and existing datasets, including the SUIM dataset [38], the WebUOT-1M dataset [41], and the COU dataset [39]. The SUIM dataset [38] and the WebUOT-1M [41] dataset are under MIT license, and the authors of the COU dataset [39] state it is available for broad use under open-source licenses. All these licenses explicitly permit non-commercial academic use and redistribution with proper attribution. For the Internet-collected images, we only retrieved images from publicly accessible platforms (e.g., Wikimedia Commons) that allow free academic use. We compare image hashes for all images of the UAM dataset, removing the overlapping samples that appeared in multiple sources. In addition, we perform source-based stratified partitioning. Images are grouped by their original collection source (e.g., specific underwater video sequences, public image repositories). Entire groups are randomly assigned to either the training set or the test set, ensuring that images from the same source, same video sequence, or same acquisition session never appear in both partitions. This prevents leakage caused by highly correlated images of the same scene or object captured in consecutive frames. A standardized pixel-level semantic segmentation annotation protocol is developed and implemented throughout the dataset construction process. First, all images are converting to the standard RGB color space to ensure annotation consistency. Subsequently, the SAM 3 model [42] is employed to generate automatic instance segmentation masks for all target objects, providing a high-quality starting point for manual refinement. Next, one trained annotator independently refines the automatically generated masks using the LabelMe tool, with specific instructions to correct edge inaccuracies, resolve occlusion relationships between objects, and exclude irrelevant regions. Finally, 10% of the final annotations are randomly selected for a second round of independent verification to ensure that the overall annotation quality of the entire dataset meets rigorous academic research standards.

A few samples from the UAM dataset are shown in Fig. 3. We consider three object classes for semantic labeling: i.e., *nature* (including waterbody/animal/plant/seafloor/wreck), *human* (including diver/swimmer), *vehicle* (including AUV/ROV/ship/submarine). The number of images covering different object classes is shown in Fig. 4. Following the standard convention adopted by widely used classic computer vision benchmarks including MNIST, CIFAR-10, CIFAR-100, and Fashion-MNIST, the UAM dataset does not include an official validation split but instead allow researchers to derive one from the training set according to their specific

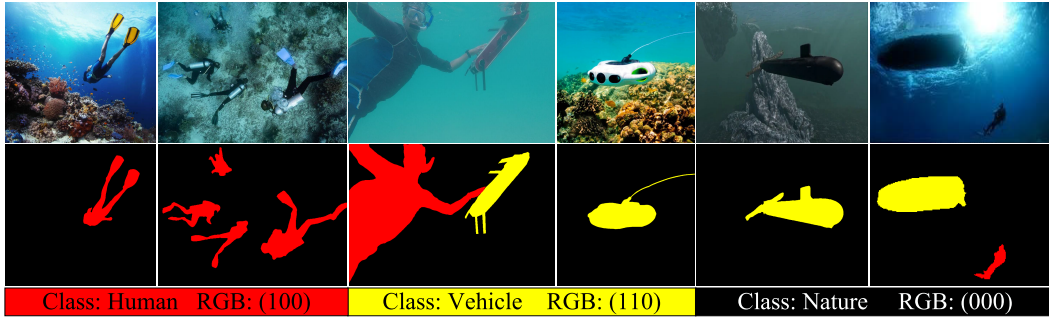


Fig. 3. Example images and corresponding pixel-annotations from the UAM dataset.

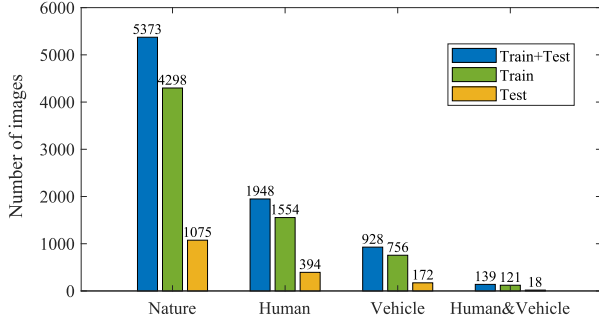


Fig. 4. The number of images covering object classes.

experimental needs, offering maximum flexibility for different tasks and model architectures. For the experiments in this paper, we split the 4298-image training set into 3868 training images and 430 validation images using a 9:1 stratified sampling ratio.

B. Scheme Design for Covert Semantic Communication

The covert semantic communication scheme in a W2A-OWC system is shown in Fig. 5. The legitimate transmitter Alice incorporates modules including semantic representation, semantic encoding, and sparse mask generator. To address the inherent challenges of underwater datasets including blurring, color distortion, uneven illumination, and turbidity, we implement a comprehensive data augmentation pipeline during the training phase. This pipeline combines geometric transformation and color space enhancement techniques to balance data diversity and training stability. The geometric transformations include random scaling, random translation, random shearing, random horizontal/vertical flipping, and random rotation, which enhance the model's robustness to variations in target scale, position, and orientation. The color space enhancements include random brightness/contrast adjustment and random gamma correction, which effectively mitigate underwater color cast and uneven lighting artifacts. In the semantic representation module, the i -th input image I^i with dimension $(3, H, W)$ is first fed into the ResNet50 backbone network to extract feature maps at $1/4$, $1/8$, $1/16$, and $1/32$ scales, where 3 represents the number of RGB channels, H is the height of image, and W is the width. These feature maps are then fused via the feature pyramid network (FPN) to integrate multi-scale information,

addressing the poor adaptability of single-scale features to both large and small target segmentation. The fused feature map is upsampled to the original image resolution, and finally processed by the seg-head network to output the semantic segmentation result J^i with dimension (C, H, W) , where C is the number of object classes.

The extracted semantic information J^i is encoded by the semantic encoding module. In this module, ResNet18 performs step-by-step downsampling to reduce the feature scale. The fused features after FPN is further compressed by the down-sampler network, yielding a tensor with dimension $(N_t, 1, 1)$, which is then flattened into a compact latent vector e^i with blocklength N_t . To increase the sparsity of latent vector for covertness, we design the sparse mask generator module, whose inputs include the latent vector e^i and $\gamma = 1/N_b$ within one batch, where N_b is the noise variance. This module can adaptively adjust the sparsity ratio of the generated sparse mask m^i according to the channel noise, thereby satisfying the dual requirements of covertness and semantic accuracy. During training, we generate a hard binary mask $m^i \in \{0, 1\}^{N_t}$ by sampling from a Bernoulli distribution parameterized by the continuous sigmoid mask $m_c^i \in [0, 1]^{N_t}$, i.e., $m^i = \mathbb{I}(m_c^i \geq \varpi)$, where $\varpi \sim \text{Uniform}(0, 1)^{N_t}$ is the sampling random vector, $\mathbb{I}(\cdot)$ is the indicator function that outputs 1 if the condition is true and 0 otherwise. The fixed threshold of 0.5 is not adopted in the training phase, since it would introduce severe training instability around boundary values such as 0.49 and 0.51. A minuscule, possibly noisy, fluctuation in the continuous vector m_c^i could abruptly flip the mask state (from 0 to 1 or vice versa), disrupting the smooth optimization of latent feature selection and potentially trapping the model in poor local optima. Then, we adopt the straight-through estimator (STE) to bypass the non-differentiable decision operation and propagate gradients directly through m_c^i during backpropagation [43]. During inference, we use a fixed threshold of 0.5 to generate deterministic binary masks for consistent performance, i.e., $m^i = \mathbb{I}(m_c^i \geq 0.5)$. Then, the sparse mask m^i is multiplied element-wise with the latent vector from semantic encoding to form the vector x^i for channel transmission, which retains only the dimensions of the latent vector critical to semantics. For zero-valued symbols in the transmitted vector, the corresponding time slot is reserved but left idle with no transmission.

After passing through E/O conversion, the W2A-OWC channel, and O/E conversion, the received vector y^i at the legitimate receiver Bob is sent into the semantic decoding

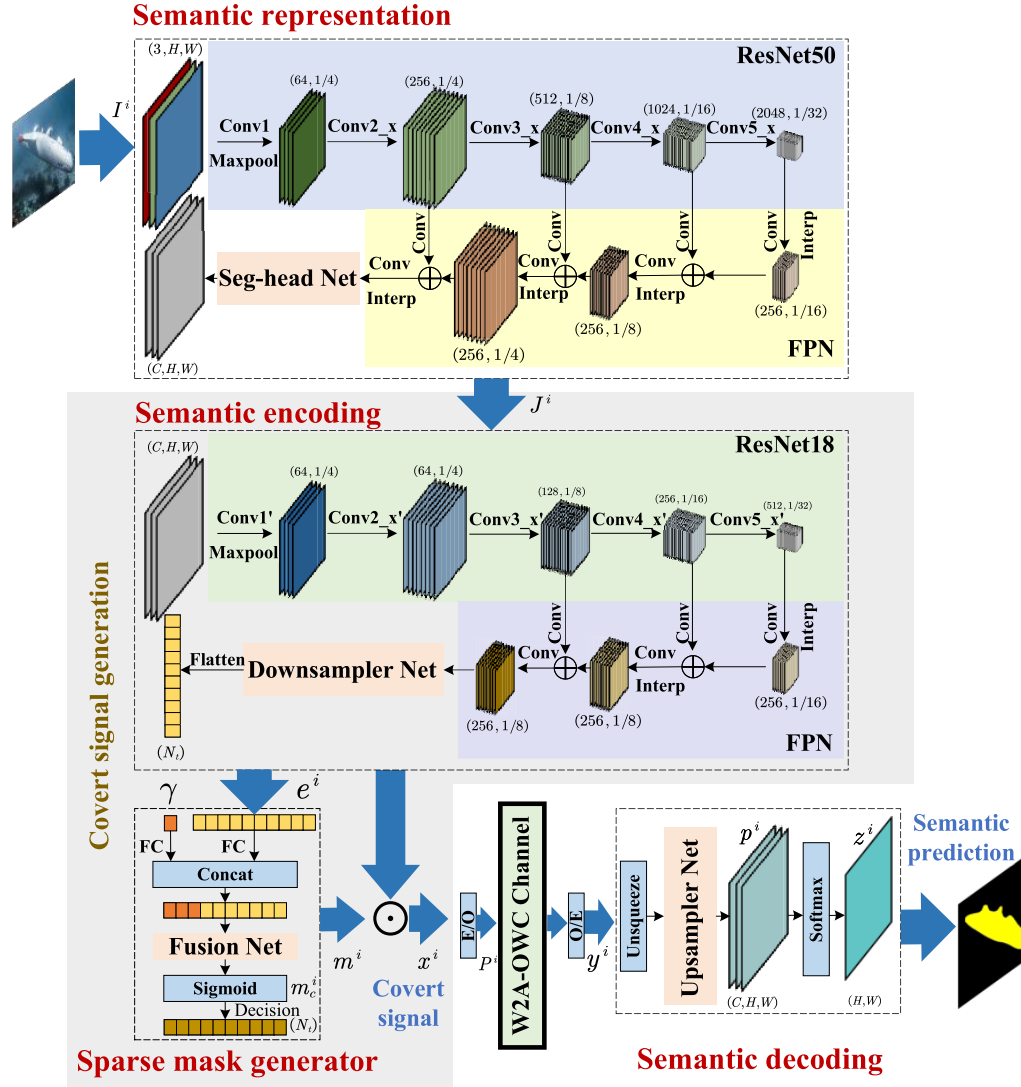


Fig. 5. Scheme structure for covert semantic communication.

module. The vector is first restored to a tensor with three dimensions $(N_t, 1, 1)$ via an unsqueeze operation, then the step-by-step upsampling restores the low-dimensional latent vector to the original image resolution, recovering the spatial information of semantic segmentation and yielding the semantic information p^i with dimension (C, H, W) . Finally, a softmax activation outputs the pixel-level semantic prediction result z^i with dimension (H, W) . Considering the sparsity constraint on m^i , the loss function between semantic information p^i and ground truth labels g^i from UAM dataset is $L = \frac{1}{B} \sum_{i=1}^B f_{\text{CEloss}}(p^i, g^i) + \lambda (\gamma \|m^i\|_1)$, where B is batchsize, f_{CEloss} is the cross-entropy loss function, λ is the weight, and $\|\cdot\|_1$ represents the L1 norm. The loss function comprises a pixel-level cross-entropy term and an L1 regularization term: the former ensures semantic transmission fidelity for underwater anomaly detection, while the latter induces sparsity by driving sparse mask elements toward zero during backpropagation, which elevates the encoding vector sparsity ratio and consequently reduces the KL divergence to enhance

signal covertness, as shown in Section II-B. The regularization coefficient γ is dynamically assigned as the reciprocal of the channel noise variance, facilitating adaptive sparsity tuning, i.e., it diminishes under high-noise conditions to relax regularization and preserve more non-zero symbols for semantic accuracy, and amplifies under low-noise conditions to intensify regularization and prune redundant semantic dimensions for improved covertness.

IV. SIMULATION RESULTS

The input images from the UAM dataset are normalized to a size of $H = 224$ and $W = 224$. The network is trained for a total of 200 epochs with parameters including batchsize $B = 10$, the number of object classes $C = 3$, encoded blocklength $N_t = 2048$, $\lambda = 1 \times 10^{-3}$, and the initial learning rate 1×10^{-4} . The W2A-OWC channel gain $h_b \sim \text{Lognormal}(\mu_b = -13.15, \sigma_b = 0.33)$ is calculated from the theoretical model in [31] when Alice and Bob locate at $(0, 0, -2)$ m and $(0, 0, 1)$ m, respectively, the water type is

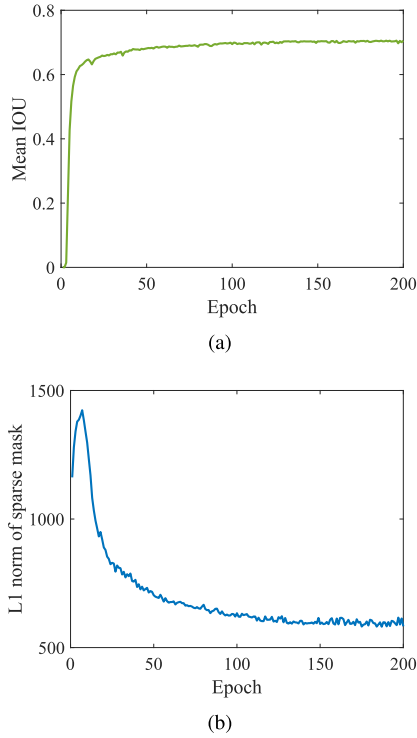


Fig. 6. (a) The mean IOU and (b) the L1 norm of sparse mask versus epoch.

Jerlov IB, as well as the wave slope follows truncated logistic distribution with parameter $\mu_s = 0$ and $\sigma_s = 5$. Similarly, the distribution parameters of Willie's channel gains h_w at different locations can be derived from the theoretical model. The non-zero-symbol energy of transmitted encoding vector x^i is normalized to 1. Considering the low optical power in covert transmission and the dominance of thermal noise, we randomly vary the noise variance within the range of -10 dBm to -25 dBm (relative to the signal power of 1 mW) in each batch to enhance the generalization of semantic communication performance under different channel quality conditions. We adopt the Adam optimizer with a CosineAnnealingLR learning rate scheduler. We calculate the semantic transmission accuracy between prediction results z^i and ground truth labels g^i based on the standard metric, i.e., the mean intersection over union (IOU) score [44]. The experiments are conducted on a single NVIDIA GeForce RTX 5080.

During training process, it is seen from Fig. 6(a) that the mean IOU of anomalous objects (vehicle and human) increases from 0 to 0.7053 as the epoch increases from 1 to 200. It indicates that the semantic transmission accuracy is gradually improved with the training process. The L1 norm of sparse mask increases from 1162 to 1423 as epoch increases from 1 to 7 and then decreases to 582 at epoch 200, as shown in Fig. 6(b). The reason is that the network first adjusts the non-zero symbol number in transmission encoding vectors based on the input image, and then gradually filters out non-core or redundant semantics to achieve sparsification of the transmitted symbols.

We test the covertness-oriented semantic communication scheme under different noise power, as shown in Fig. 7. In

Fig. 7(a), the mean IOU of anomalous objects (vehicle and human) with sparse mask decreases from 0.7057 to 0.6962 as noise power increases from -25 dBm to -10 dBm, which is close to that without sparse mask. To isolate the effect of semantic sparse selection from the effect of using less total energy, we adopt an energy-matched baseline without sparse mask, which rescales its dense latent vector on a per-sample basis so that its total transmit energy is exactly equal to that of the proposed sparse signal. It shows that the mean IOU of the energy-matched baseline is lower than that of sparse-mask-based scheme at all noise powers. It is seen from Fig. 7(b) that the sparsity ratio of the encoding vectors with sparse mask decreases from 0.7164 to 0.6987 as noise power increases from -25 dBm to -10 dBm. Since higher noise power benefits signal covertness but degrades semantic transmission accuracy, the parameter γ in our designed loss function achieves adaptive adjustment of the sparsity ratio of transmitted vectors under different noise power levels. Specifically, when the noise power is high, as it can effectively conceal the transmitted signals, we can reduce the sparsity ratio of transmitted vectors to maximize semantic transmission accuracy; conversely, when the noise power is low, we can increase the sparsity ratio of transmitted vectors to satisfy the covertness requirement. Compared with the case without sparse mask, it indicates that the proposed scheme adaptively reduces the number of non-zero symbols in accordance with the noise power variation while preserving the accuracy of semantic information. In Fig. 7(c), when Willie is at (0, 0, 1) m, average KL divergence of Willie across different block fading coefficients with sparse mask decreases from 0.5675 to 0.1284 as noise power increases from -25 dBm to -10 dBm, since strong noise facilitates signal hiding. The average KL divergence in the case without sparse mask is much higher than that with the sparse mask, and the ratio between the two cases reaches a maximum of 3.38, which verifies the effectiveness of the proposed scheme in enhancing signal covertness. In addition, the proposed scheme with sparse mask still achieves substantially lower KL divergence than the energy-matched baseline without sparse mask across all noise powers. For example, at -20 dBm, the proposed scheme obtains a KL divergence of 0.3569, whereas the energy-matched baseline gives 0.7621 under the same total transmit energy. These results confirm that the covertness improvement is not only a consequence of reduced transmit energy/rate. Rather, the learned sparse mask selectively preserves task-relevant semantic dimensions while suppressing redundant or highly observable latent components, which reduces Willie's statistical distinguishability even when the baseline is constrained to the same total transmit energy. We further evaluated the covertness of the proposed scheme with sparse mask when Willie is located at different positions. As shown in Fig. 7(d), when Willie moves farther away from the legitimate receiver, the average KL divergence is significantly reduced, which decreases from 0.3181 to 0.0126 at -20 dBm when Willie's height increases from 1 m to 5 m. The reason is that the Willie-side channel gain becomes weaker with distance, so the received signal component is more severely attenuated and becomes closer to the noise-only observation. These results show that the proposed scheme

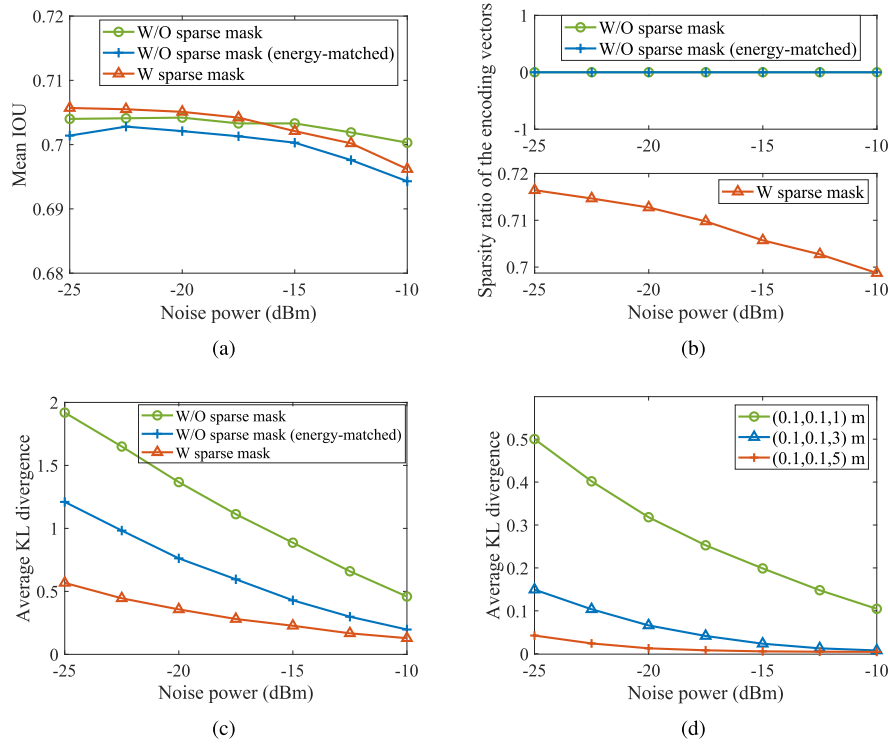


Fig. 7. (a) The mean IOU of Bob, (b) the sparsity ratio of the encoding vectors, (c) the average KL divergence at Willie's position of (0, 0, 1) m, and (d) the average KL divergence at different Willie's positions versus noise power (relative to the signal power of 1 mW) in underwater anomaly monitoring tasks.

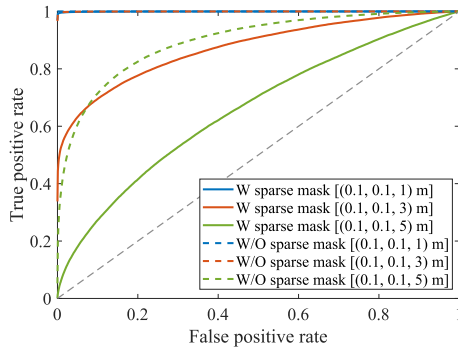


Fig. 8. ROC curves of Willie's energy detector (noise power of -10 dBm).

maintains low statistical detectability, especially when Willie is not located near the legitimate receiver, demonstrating its robustness for covert semantic communication.

To further evaluate the proposed scheme under a concrete detector, we consider an energy detector at Willie, where the test statistic is defined as $T(\mathbf{y}_w) = \sum_{i=1}^{2048} y_w(i)^2 / 2048$. For each Willie position, we repeat the channel and noise realizations 100 times for every test image and obtain the receiver operating characteristic (ROC) curves shown in Fig. 8, where noise power is considered as -10 dBm. Compared with the baseline without sparse mask, the proposed scheme consistently reduces Willie's detection probability (i.e., true positive rate) under the same false alarm probability (i.e., false positive rate). For example, when Willie is located at (0.1, 0.1, 3) m and the false alarm probability is 0.01, the detection probability of the baseline without sparse mask is

0.9968, whereas that of the proposed scheme is reduced to 0.5315. When Willie moves farther away to (0.1, 0.1, 5) m, the detection probability at the same false alarm probability decreases from 0.3862 for the baseline without sparse mask to only 0.0668 for the proposed scheme. When Willie is close to Bob, e.g., at (0.1, 0.1, 1) m, both schemes are relatively easy to detect because Willie experiences a strong observation channel. However, such a near-coincident Willie-Bob geometry is less representative of practical monitoring scenarios. Overall, the sparse-mask design substantially pushes Willie's ROC curve toward the random-guessing diagonal under weaker Willie-side channels. These ROC results are also consistent with the KL-divergence analysis, supporting that the proposed scheme lowers Willie's statistical distinguishability.

The performance of the proposed scheme under different water types is shown in Fig. 9. The mean IOU under Jerlov IB type water is higher than that under the Jerlov II and IC type water, and the average KL divergence under Jerlov IB type water is also higher than the other two water types. Since the total attenuation coefficient of water at wavelength 525 nm increases in the following order: Jerlov Type IB, Jerlov Type II, Jerlov Type IC [45], and a more attenuated water channel improves covertness by suppressing the observable semantic signal, but reduces the received signal strength, thereby degrading the semantic accuracy. It indicates that when transmitting in more turbid water bodies, the transmit power can be appropriately increased to ensure semantic transmission accuracy, while the trade-off with KL divergence must also be considered.

To verify the necessity of the semantic encoding and decoding modules in the proposed scheme, we perform an

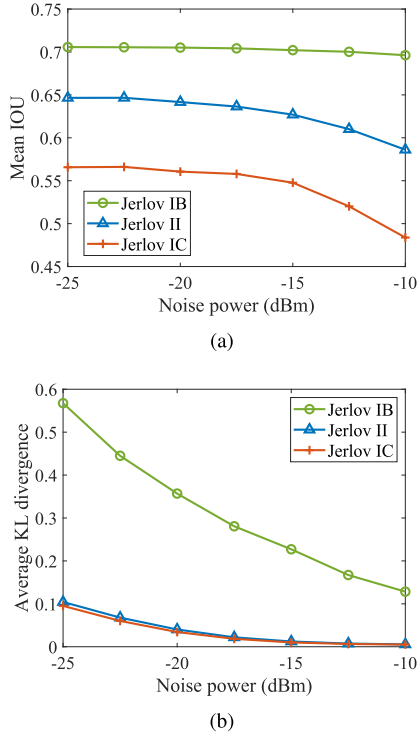


Fig. 9. (a) The mean IOU of Bob, and (b) the average KL divergence at Willie's position of (0, 0, 1) m under different water types versus noise power (relative to the signal power of 1 mW) in underwater anomaly monitoring tasks.

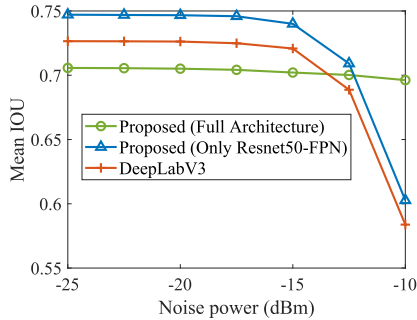


Fig. 10. The performance comparison of proposed complete scheme, segmentation-only transmission scheme and DeepLabV3 scheme.

ablation study in which only the ResNet50-FPN semantic representation module is retained. Its output semantic probability map is normalized to unit average symbol energy and transmitted over the W2A-OWC channel. At low noise power in Fig. 10, directly transmitting the dense ResNet50-FPN probability map can achieve slightly higher mean IoU than the proposed full architecture, e.g., 0.7471 at -25 dBm, because it transmits the entire dense segmentation probability with 150528 symbols per image. However, this comes at a much larger transmission cost and energy. The proposed scheme transmits only a 2048-dimensional latent vector and further activates only a sparse subset of semantic symbols, and maintains stable mean IOU around 0.70 across the noise power range. At high noise power, the proposed scheme remains more robust and the mean IOU of ResNet50-FPN-only scheme drops to 0.6027. This is because the proposed

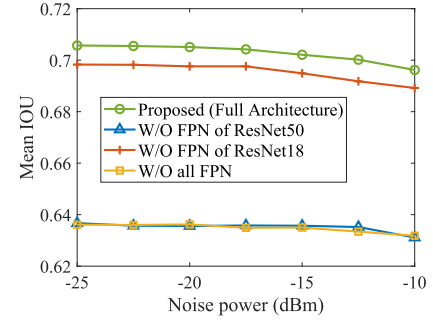


Fig. 11. FPN ablation results of the proposed scheme.

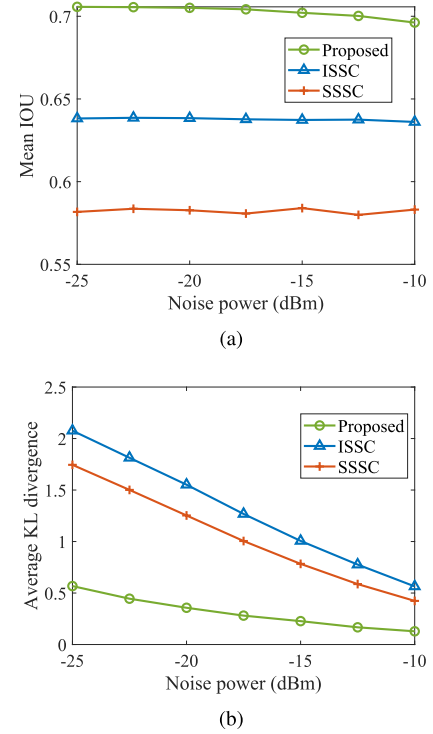


Fig. 12. (a) The mean IOU of Bob and (b) the average KL divergence at Willie's position of (0, 0, 1) m in different semantic communication schemes.

latent representation is not a simple downsampled probability map. It is learned jointly with the channel-aware semantic decoder. The sparse mask generator selects the most informative semantic symbols, and the receiver reconstructs the segmentation through a trained semantic decoder rather than a pixel-wise argmax operation. Therefore, although fewer symbols are transmitted, each active symbol carries more compact semantic information and the receiver has learned to suppress channel-induced perturbations. We also observe that the ResNet50-FPN semantic segmentation module used in our framework is stronger than the typical DeepLabV3 baseline in [46] on this underwater anomaly monitoring task. Under the same direct semantic-map transmission setting, ResNet50-FPN consistently achieves higher anomalous-object mean IOU than DeepLabV3 over all noise powers.

To investigate the effects of FPN on segmentation accuracy, we conduct an FPN ablation study, as shown in Fig. 11. It

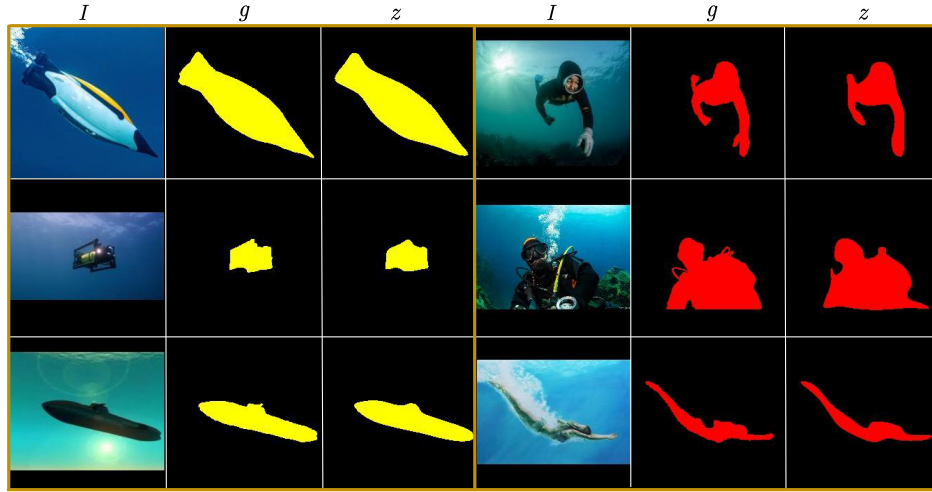


Fig. 13. A few sample images I , corresponding ground truth labels g , and semantic prediction results z .

is seen that the complete scheme consistently achieves the best mean IOU over all noise power. Removing the FPN of ResNet50 causes a clear performance drop, with the mean IOU decreasing to 0.6356 at -20 dBm. Removing all FPN modules shows a similar degradation. This indicates that the FPN in the ResNet50 segmentation module is important for preserving multi-scale spatial semantics and small anomalous-object boundaries in underwater scenes. Removing the FPN of ResNet18 in the semantic encoding module leads to a smaller but still observable decrease, with the mean IOU dropping from 0.7051 to 0.6976 at -20 dBm. This suggests that the encoder-side FPN also contributes to more robust semantic feature compression before channel transmission. These results demonstrate that the FPN design improves the quality of both semantic representation and semantic encoding, enabling the proposed scheme to maintain more accurate and robust anomalous-object segmentation.

Figure 12 shows a performance comparison between the proposed scheme and two existing image-segmentation-oriented semantic communication baselines. The image segmentation semantic communication (ISSC) baseline in [47] is implemented by adapting the ISSC idea to the underwater semantic segmentation task, where Swin-based multi-scale semantic features are compressed into a 2048-dimensional latent vector, transmitted through the W2A-OWC channel, and decoded into segmentation logits. The semantic segmentation-based semantic communication (SSSC) baseline in [48] is achieved by adapting the regions of interest (ROI)/ regions of non-interest (RONI) semantic transmission mechanism to direct segmentation, where the ROI and RONI semantic branches share the same total 2048-dimensional transmitted latent vector. In terms of segmentation accuracy in Fig. 12(a), our proposed scheme maintains stable anomalous-object mean IOU across all noise power, around 0.70, and clearly outperforms the ISSC scheme and the SSSC scheme. In Fig. 12(b), the proposed scheme reduces the average KL divergence to 0.3569, compared with 1.5518 for the ISSC scheme and 1.2531 for the SSSC scheme at noise power of -20 dBm. This corresponds to approximately 77% lower KL divergence than

ISSC and 71% lower KL divergence than SSSC. These results demonstrate that the proposed scheme is more suitable for covert semantic communication and provides better semantic accuracy and signal covertness in underwater anomaly monitoring tasks.

The qualitative comparison results under noise power of -20 dBm for the proposed scheme are visualized in Fig. 13, which shows that the semantic prediction results z are close to the ground truth labels g .

V. CONCLUSION

In this work, we have proposed a covert semantic transmission scheme for W2A-OWC systems. To explore the covert signal design, the analytical relationship between the sparsity ratio of transmitted encoding vectors and covert performance metrics has been obtained through theoretical derivation. The semantic representation, semantic encoding, semantic decoding, and sparse mask generator modules have been jointly developed and optimized to enhance signal covertness while ensuring the semantic transmission accuracy. For the underwater monitoring task, the UAM dataset containing a variety of underwater anomalous objects has been constructed for the first time. Simulation results show that the proposed scheme achieves high semantic transmission accuracy and enables adaptive adjustment of the sparsity ratio across diverse channel noise scenarios. The scheme with sparse masks reduces the average KL divergence by a factor of up to 3.38, leading to a significant improvement in signal covertness. A limitation of the present work is that it focuses on data-driven and model-based validation of the proposed semantic communication scheme, rather than a new physical W2A-OWC hardware demonstration. Hardware-in-the-loop testing remains necessary for deployment-oriented validation. Therefore, future work will involve implementing the proposed scheme on underwater AUV and aerial UAV platforms, establishing a W2A-OWC experimental link in natural waterbody environments, and evaluating the effectiveness and covertness of the scheme in practical underwater anomaly monitoring tasks.

REFERENCES

- [1] H. Luo, J. Wang, F. Bu, R. Ruby, K. Wu, and Z. Guo, "Recent progress of air/water cross-boundary communications for underwater sensor networks: A review," *IEEE Sensors J.*, vol. 22, no. 9, pp. 8360–8382, May 2022.
- [2] L.-K. Chen, Y. Shao, and Y. Di, "Underwater and water-air optical wireless communication," *J. Lightw. Technol.*, vol. 40, no. 5, pp. 1440–1452, Mar. 15, 2022.
- [3] T. Lin et al., "Waving effect characterization for water-to-air optical wireless communication," *J. Lightw. Technol.*, vol. 41, no. 1, pp. 120–136, Jan. 15, 2023.
- [4] X. Chen et al., "Covert communications: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 1173–1198, 2nd Quart., 2023.
- [5] H. Zhu, H. Wu, and X. Jiang, "Jamming with zero-forcing beamforming for covert communication in MIMO systems," in *Proc. Int. Conf. Netw. Neww. Appl. (NaNA)*, Aug. 2023, pp. 122–126.
- [6] R. Ma, W. Yang, X. Guan, X. Lu, Y. Song, and D. Chen, "Covert mmWave communications with finite blocklength against spatially random wardens," *IEEE Internet Things J.*, vol. 11, no. 2, pp. 3402–3416, Jan. 2024.
- [7] Y. Zhang, Y. Zhang, J. Wang, S. Xiao, W. Ni, and W. Tang, "Robust beamfocusing for FDA-aided near-field covert communications with uncertain location," in *Proc. IEEE Int. Conf. Commun. Workshops*, May 2023, pp. 1349–1354.
- [8] Z. Duan, X. Yang, Y. Gong, D. Wang, and L. Wang, "Covert communication in uplink NOMA systems under channel distribution information uncertainty," *IEEE Commun. Lett.*, vol. 27, no. 5, pp. 1282–1286, May 2023.
- [9] L. Huang, J. Lei, and Y. Huang, "Spatial modulation covert communication assisted by artificial noise," in *Proc. IEEE/CIC Int. Conf. Commun. China*, Aug. 2023, pp. 1–6.
- [10] Z. Gao, H. Li, Y. Ye, and G. Lu, "Intermittent artificial noise assisted covert cooperative backscatter communication with residual hardware impairments," *IEEE Trans. Green Commun. Netw.*, vol. 9, no. 3, pp. 817–828, Sep. 2025.
- [11] F. Shu, T. Xu, J. Hu, and S. Yan, "Delay-constrained covert communications with a full-duplex receiver," *IEEE Wireless Commun. Lett.*, vol. 8, no. 3, pp. 813–816, Jun. 2019.
- [12] H. Chang, X. Kang, H. Lei, T. A. Tsiftsis, G. Pan, and H. Liu, "STAR-RIS-aided covert communications in MISO-RSMA systems," *IEEE Trans. Green Commun. Netw.*, vol. 8, no. 4, pp. 1318–1331, Dec. 2024.
- [13] B. Che, C. Gao, R. Ma, X. Zheng, and W. Yang, "Covert wireless communication in multichannel systems," *IEEE Wireless Commun. Lett.*, vol. 11, no. 9, pp. 1790–1794, Sep. 2022.
- [14] R. He et al., "When the warden does not know transmit power: Detection performance analysis and covertness strategy design," *IEEE Trans. Commun.*, vol. 72, no. 6, pp. 3184–3198, Jun. 2024.
- [15] W. Zhang et al., "Joint trajectory and beamforming optimization for IRS-assisted multi-antenna UAV covert communications with a finite blocklength," *IEEE Trans. Green Commun. Netw.*, vol. 10, pp. 426–439, 2026.
- [16] Q. Hu, T. Lin, T. Wei, N. Huang, Y.-J. Zhu, and C. Gong, "Covert transmission in water-to-air optical wireless communication systems," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 4432–4447, 2024.
- [17] X. Luo, H.-H. Chen, and Q. Guo, "Semantic communications: Overview, open issues, and future research directions," *IEEE Wireless Commun.*, vol. 29, no. 1, pp. 210–219, Feb. 2022.
- [18] W. Yang et al., "Semantic communications for future internet: Fundamentals, applications, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 213–250, 1st Quart., 2023.
- [19] Y. Yang et al., "Semantic communications with artificial intelligence tasks: Reducing bandwidth requirements and improving artificial intelligence task performance," *IEEE Ind. Electron. Mag.*, vol. 17, no. 3, pp. 4–13, May 2022.
- [20] J. Hu, L. Ye, Y. Chen, X. Zhang, J. Wang, and Z. Chen, "Covert communications for text semantic with finite blocklength," *IEEE Wireless Commun. Lett.*, vol. 13, no. 10, pp. 2842–2846, Oct. 2024.
- [21] R. Xu, G. Li, Z. Yang, M. Chen, Y. Liu, and J. Li, "Covert and reliable semantic communication against cross-layer privacy inference over wireless edge networks," in *Proc. IEEE Conf. Wireless Commun. Netw. (WCNC)*, Apr. 2024, pp. 1–6.
- [22] H. Xie, Z. Qin, G. Y. Li, and B.-H. Juang, "Deep learning enabled semantic communication systems," *IEEE Trans. Signal Process.*, vol. 69, pp. 2663–2675, 2021.
- [23] Q. Hu, G. Zhang, Z. Qin, Y. Cai, G. Yu, and G. Y. Li, "Robust semantic communications with masked VQ-VAE enabled codebook," *IEEE Trans. Wireless Commun.*, vol. 22, no. 12, pp. 8707–8722, Dec. 2023.
- [24] Y. Wang, Y. Hu, H. Du, T. Luo, and D. Niyato, "Multi-agent reinforcement learning for covert semantic communications over wireless networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.
- [25] W. Zhang, Y. Hu, T. Luo, Z. Zhang, and M. Chen, "Optimization of private semantic communication performance: An uncooperative covert communication method," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 1408–1421, 2026.
- [26] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [27] K. Tang, Y. Niu, J. Huang, J. Shi, and H. Zhang, "Unbiased scene graph generation from biased training," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3716–3725.
- [28] J. Xu, Z. Huang, W. Zhai, H. Qiu, Y. Gao, and Y. Ji, "Environment semantics aided underwater wireless optical semantic communication," *J. Lightw. Technol.*, vol. 43, no. 3, pp. 1186–1202, Feb. 15, 2025.
- [29] J. Xu, Z. Huang, Y. Gao, H. Qiu, and Y. Ji, "Adaptive symbol and power loading for OFDM-based underwater wireless optical semantic communications," *IEEE Internet Things J.*, vol. 13, no. 8, pp. 17643–17657, Apr. 2026.
- [30] J. Xu, Z. Huang, Y. Gao, W. Zhai, H. Qiu, and Y. Ji, "Design and experimental demonstration of underwater wireless optical communication system based on semantic communication paradigm," *Opt. Exp.*, vol. 32, no. 2, pp. 2188–2201, Jan. 2024.
- [31] Q. Hu, C. Gong, T. Lin, J. Luo, and Z. Xu, "Secrecy performance analysis for water-to-air visible light communication," *J. Lightw. Technol.*, vol. 40, no. 14, pp. 4607–4620, Jul. 15, 2022.
- [32] F. M. Davidson and X. Sun, "Gaussian approximation versus nearly exact performance analysis of optical communication systems with PPM signaling and APD receivers," *IEEE Trans. Commun.*, vol. 36, no. 11, pp. 1185–1192, Nov. 1988.
- [33] J.-B. Wang, Q.-S. Hu, J. Wang, M. Chen, and J.-Y. Wang, "Tight bounds on channel capacity for dimmable visible light communications," *J. Lightw. Technol.*, vol. 31, no. 23, pp. 3771–3779, Dec. 15, 2013.
- [34] B. A. Bash, D. Goekel, and D. Towsley, "Limits of reliable communication with low probability of detection on AWGN channels," *IEEE J. Sel. Areas Commun.*, vol. 31, no. 9, pp. 1921–1930, Sep. 2013.
- [35] T. M. Cover, *Elements of Information Theory*. Hoboken, NJ, USA: Wiley, 1999.
- [36] C. Cachin, "An information-theoretic model for steganography," *Inf. Comput.*, vol. 192, no. 1, pp. 41–56, Jul. 2004.
- [37] S. Yan, B. He, X. Zhou, Y. Cong, and A. L. Swindlehurst, "Delay-intolerant covert communications with either fixed or random transmit power," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 1, pp. 129–140, Jan. 2019.
- [38] M. J. Islam et al., "Semantic segmentation of underwater imagery: Dataset and benchmark," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2020, pp. 1769–1776.
- [39] R. Mukherjee, S. Singh, J. McWilliams, and J. Sattar, "The common objects underwater (COU) dataset for robust underwater object detection," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2025, pp. 18597–18603.
- [40] M. Waszak, A. Cardaillac, B. Elvessæter, F. Rødølen, and M. Ludvigsen, "Semantic segmentation in underwater ship inspections: Benchmark and data set," *IEEE J. Ocean. Eng.*, vol. 48, no. 2, pp. 462–473, Apr. 2023.
- [41] C. Zhang, L. Liu, G. Huang, H. Wen, X. Zhou, and Y. Wang, "WebUOT-1M: Advancing deep underwater object tracking with a million-scale benchmark," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2024, pp. 50152–50167.
- [42] N. Carion et al., "SAM 3: Segment anything with concepts," Nov. 2025, *arXiv:2511.16719*.
- [43] Y. Bengio, N. Léonard, and A. Courville, "Estimating or propagating gradients through stochastic neurons for conditional computation," Aug. 2013, *arXiv:1308.3432*.
- [44] S. Nowozin, "Optimal decisions from probabilistic models: The intersection-over-union case," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 548–555.
- [45] M. G. Solonenko and C. D. Mobley, "Inherent optical properties of Jerlov water types," *Appl. Opt.*, vol. 54, no. 17, pp. 5392–5401, Jun. 2015.
- [46] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," Aug. 2017, *arXiv:1706.05587*.

- [47] Q. Pan et al., "Image segmentation semantic communication over Internet of Vehicles," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Mar. 2023, pp. 1–6.
- [48] J. Wu et al., "Semantic segmentation-based semantic communication system for image transmission," *Digit. Commun. Netw.*, vol. 10, no. 3, pp. 519–527, Jun. 2024.



Qingqing Hu (Member, IEEE) received the B.S. degree from Dalian University of Technology, Dalian, China, in 2016, and the M.S. and Ph.D. degrees from the University of Science and Technology of China, Hefei, China, in 2019 and 2024, respectively. She has been a Lecturer with the School of Computer and Information, Hefei University of Technology, Hefei, since 2024. She is currently a Post-Doctoral Researcher with New York University Abu Dhabi (NYUAD). Her research interests include information security, signal processing, and optical wireless communication.



Bin Chen (Senior Member, IEEE) was born in Hefei, China. He received the B.Sc. degree in electronic information science and technology from Hefei University of Technology, Hefei, in 2010, and the Ph.D. degree in electronics and communications engineering from the University College Dublin, Dublin, Ireland, in 2015. From 2016 to 2019, he was a Post-Doctoral Researcher with the Signal Processing Systems Group and the Electro-Optical Communication Systems Group, Department of Electrical Engineering, Eindhoven University of

Technology, Eindhoven, The Netherlands. He is currently a Professor with Hefei University of Technology. He is the author or co-author of more than 100 technical publications and holds more than ten patents. His research interests include developing practical and novel schemes for digital communication systems related to optical communications, coded modulation, information theory, channel coding, and signal processing. He was a recipient of multiple awards, including Asia Communications and Photonics Conference (ACP2018) Best Paper Award, the Optoelectronics and Communications Conference (OECC2019) Best Paper Award, Asia Communications and Photonics Conference (ACP2022) Young Scientist Award, and European Conference on Optical Communication (ECOC2024) Top-Scored Paper. He also served on Technical Program Committees and he was the Workshop/the Session Co-Chair for several international conferences, including OFC, OECC, and ACP.



Yi Lei was born in Chongqing, China. She received the B.E. degree in electronic information and technology program from Beijing Forestry University (BJFU), Beijing, China, in 2013, and the Ph.D. degree in electronic science and technology from Beijing University of Posts and Telecommunications (BUPT), Beijing, in 2019. She held a visiting Ph.D. position with the Signal Processing Systems (SPS) Group, Department of Electrical Engineering, Eindhoven University of Technology (TU/e), Eindhoven, The Netherlands, from August 2017 to August 2018. She is currently an Associate Professor with Hefei University of Technology, Hefei, China. Her research interests include optical communication systems, forward error correction, coded modulation, and physical layer security.



Chen Gong (Senior Member, IEEE) received the B.S. degree in electrical engineering and mathematics (minor) from Shanghai Jiaotong University, Shanghai, China, in 2005, the M.S. degree in electrical engineering from Tsinghua University, Beijing, China, in 2008, and the Ph.D. degree from Columbia University, New York City, NY, USA, in 2012. He was a Senior Systems Engineer with Qualcomm Research, San Diego, CA, USA, from 2012 to 2013. He is currently a Faculty Member with the University of Science and Technology of China.

His research interests include wireless communications, optical wireless communications, and signal processing. He was selected by the Young 1000 Talent Program of China Government in 2014. He was awarded by Hongkong Qishi Outstanding Young Researcher Award in 2016.